Module 04: Cognition — Thinking With and Through the Body

Learning Objectives

1. Explain the concept of **embodied cognition** as it relates to the felt sense — understanding that thinking is not disembodied computation but a somatically grounded process.
2. Describe Mark Johnson's theory of **image schemas** and **embodied metaphor** as evidence that abstract thought arises from bodily experience.
3. Analyze how **interoceptive signals** contribute to cognitive processes including decision-making, judgment, and reasoning.
4. Recognize the limitations of purely computational models of cognition and the necessity of somatic grounding for a complete account of thought.

Introduction

Consider the phrase "I feel weighed down by this problem." This is not merely a figure of speech. Research in embodied cognition reveals that holding a heavy object literally makes problems seem more serious, that physical warmth increases feelings of social warmth, and that leaning forward biases people toward future-oriented thinking. Thought, it turns out, is not the pure operation of a disembodied mind. It is drenched in the body's felt sense — shaped by posture, muscle tension, visceral state, and the somatic traces of prior experience.

The cognitive science revolution of the mid-20th century modeled the mind as a computer: input, processing, output. Embodied cognition challenges this metaphor at its foundation. Mark Johnson and George Lakoff demonstrated that our most abstract concepts — time, causation, morality, number — are structured by bodily metaphors rooted in physical experience. Andy Clark has argued that the brain is not a detached processor but a "prediction machine" that evolved to control a body, not to contemplate Platonic forms. The felt sense is the experiential proof: when we think, the body thinks with us.

Key Concepts

1. Image Schemas: The Bodily Roots of Thought

Mark Johnson (1987) identified **image schemas** — recurring patterns of bodily experience that structure abstract thought. The CONTAINER schema (inside/outside, in/out) arises from the repeated experience of being a bounded body in bounded spaces. The BALANCE schema comes from the felt experience of postural equilibrium. The SOURCE-PATH-GOAL schema emerges from the kinesthetic experience of directed movement. These schemas are not conscious concepts but pre-reflective, somatic structures that organize thinking at its deepest level. In active inference, image schemas correspond to high-level priors in the generative model, shaped by a lifetime of embodied sensorimotor experience.

2. Embodied Metaphor and Abstract Reasoning

Lakoff and Johnson (1980, 1999) showed that abstract reasoning is systematically structured by embodied metaphors. "Understanding is grasping" maps the physical act of grasping onto cognitive comprehension. "Importance is weight" maps bodily heaviness onto significance. These are not decorative linguistic choices but the very medium of abstract thought — neural circuits originally dedicated to bodily action are recruited to process abstract domains. The felt sense of understanding — the "aha" moment, the sense of cognitive "grip" — is literally the echo of the body's grasping, resonating through the cortical networks that handle both physical and conceptual hold.

3. Somatic Contributions to Decision-Making

Damasio's Iowa Gambling Task experiments demonstrated that bodily signals guide cognitive decision-making before conscious awareness. Participants developed anticipatory skin conductance responses (somatic markers) to risky decks before they could articulate which decks were dangerous. The felt sense was leading cognition, not following it. In active inference terms, interoceptive prediction errors — the body's rapid simulation of expected somatic outcomes — function as an embodied heuristic that biases deliberative processing toward viable choices.

4. Gut Feelings and Cognitive Shortcuts

The felt sense provides what Gerd Gigerenzer calls **fast and frugal heuristics** — cognitive shortcuts that bypass extensive deliberation by drawing on bodily signals. The "gut feeling" that a business deal is wrong, or that a new acquaintance is trustworthy, is not irrational mysticism. It is compressed interoceptive inference: the generative model, trained by prior experience, rapidly generating somatic predictions about expected outcomes. These predictions are experienced as feelings rather than thoughts precisely because they arise from interoceptive, not propositional, inference.

5. The Extended Mind and Felt Cognition

Andy Clark and David Chalmers's (1998) extended mind thesis proposes that cognitive processes can extend beyond the brain into the body and environment. The felt sense enriches this proposal: not only do we think with external tools (notebooks, calculators), we think with our bodies. The mathematician who paces while solving a problem, the writer who feels the rhythm of a sentence in her breathing, the architect who grasps spatial relationships through gesture — all are using the body as a cognitive instrument. The felt sense is the subjective indicator of this embodied cognitive processing.

Active Inference Connection

Active inference dissolves the traditional boundary between cognition and bodily regulation. The same generative model that predicts interoceptive states also generates cognitive predictions about abstract domains. Free energy minimization operates at all levels simultaneously: the body regulates its internal milieu, perception constructs the external world, and cognition elaborates meaning — all within a single, hierarchical predictive architecture. The felt sense during cognition — the warmth of insight, the tension of confusion, the lightness of resolution — is not cognitive noise but informative signal: the body's report on how well the generative model's cognitive predictions are faring.

Experiential Applications

- **Practice — Tracking the Felt Sense of Thinking**: While working on a moderately difficult problem (a crossword puzzle, a planning task, a personal decision), periodically pause and scan your body. Notice the felt quality of cognitive engagement: the tension in your forehead during effort, the opening in your chest when an idea emerges, the subtle nausea of confusion, the settling that accompanies resolution. These bodily states are not distractions from thinking — they are thinking, expressed in the somatic register.

- **Case Study — Embodied Mathematics**: Research by Rafael Nunez and George Lakoff has shown that mathematical concepts are grounded in embodied image schemas. The number line derives from the SOURCE-PATH-GOAL schema; set theory maps onto the CONTAINER schema; infinity extends the MORE IS UP metaphor beyond all bounds. Mathematicians frequently report "feeling" their way toward proofs — a sense of aesthetic rightness or wrongness that guides formal reasoning. This felt sense of mathematical truth is the generative model's interoceptive response to the formal coherence (or incoherence) of a mathematical structure.

Cross-References

- **Module 03 (Perception)**: Embodied perception provides the raw material on which felt cognition operates.
- **Module 05 (Action)**: Cognitive plans must ultimately be expressed through bodily action — thought and movement are continuous.
- **Unit 03 (Intuitive Knowing)**: Explores how embodied cognitive processes, over time, consolidate into tacit expertise.

Summary

Cognition is not a disembodied operation performed by a brain-computer. It is a somatically grounded process in which the body's felt sense shapes thinking at every level — from the image schemas that structure abstract concepts to the somatic markers that guide decision-making to the gut feelings that compress complex inference into rapid, embodied judgments. Active inference provides the framework: a single generative model integrates interoceptive, proprioceptive, and cognitive predictions, minimizing free energy across all domains simultaneously. When we think, the body thinks with us, and the felt sense is its voice.